

**GEOLOGY AND PETROGRAPHIC ANALYSIS OF “GRANITE-GNEISS” IN
BONGWO HILL (THREE SISTERS’ ROCK) SONG, NORTHEASTERN NIGERIA**

BY

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GLY/19U/4212

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THE AWARD OF B. Tech (HONS) DEGREE IN GEOLOGY.**

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CERTIFICATION

This is to certify that this project work titled “**Geology and petrographic analysis of Granite-Gneiss in Bongwo hills (Three sisters’ rock) Song Northeastern Nigeria**” was carried out by **Usama Aliyu Babando with Reg No (GLY/19U/4212)** in the Department of Geology, Faculty of Physical Sciences, Modibbo Adama University, Yola. This work meets the requirements for the award of the degree of **Bachelor of Technology (B.Tech) in Geology**.

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DECLARATION

I, Usama Aliyu Babando with the Registration Number GLY/19U/4212 declare that the work described in this Project represent my original work and has not been previously submitted in part or in full to any University or similar Institution for any Degree Award.

Sign: _____

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Date

DEDICATION

This work is dedicated to the Almighty Allah, who surpasses all human understanding and fulfills his words. The Lord that gives health and life, all praises, honor and adoration be unto you the maker of heaven and the earth. It is also dedicated to my beloved mother, Hajiya Zainab Aliyu Babando and my sibling especially Imran Aliyu.

ACKNOWLEDGEMENT

My sincere gratitude goes to Almighty Allah, the Most Gracious and Most Merciful, for His guidance, wisdom, and protection throughout the course of this research. I am deeply grateful to my project supervisor, **Dr. M Halilu**, for her invaluable guidance, encouragement, and constructive criticism that made this work a success. I also extend my appreciation to all lecturers and staff of the Department of Geology, Modibbo Adama University, Yola, for their support and academic contributions throughout my study.

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ABSTRACT

This study examines the geology and geochemical characteristics of the Granite–Gneiss complex in Bongwo Hill (Three Sisters' Rock), Song, Northeastern Nigeria. Field mapping, petrographic, and geochemical analyses were carried out to determine the rock assemblages and their compositional variations. The major lithological units identified include phaneritic granite, coarse- to medium-grained granite, gneiss, migmatite, basalt, and pumice. Geochemical analysis using Atomic Absorption Spectrophotometry (AAS) revealed that SiO_2 ranges from 65.80–98.70%, and Al_2O_3 from 9.70–36.50%, classifying the rocks as acidic and peralkaline granites. Harker variation diagrams show a negative correlation between SiO_2 and oxides such as Fe_2O_3 , CaO , Na_2O , and MgO , indicating fractional crystallization and separation of mafic minerals during magma evolution. The positive correlation between SiO_2 and K_2O suggests potassic enrichment at the later stage of differentiation. The granitoids are silica-rich, calc-alkaline in nature, and derived from crustal melts during the Pan-African orogeny. Structurally, the area exhibits NE–SW and NW–SE trending joints and foliations consistent with regional tectonics. The study concludes that the Bongwo Hill granites were emplaced during Pan-African magmatism and are significant sources of industrial minerals and construction materials.

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CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

The study area is situated in the Hawal Massif, Northeastern Nigeria. The Hawal Massif has undergone Pre-Pan African orogeny. This has resulted into the consequent metamorphism, migmatization and granitization of the pre-existing rocks (Rahman, 1981). Nigeria has a diverse geology, and the basement rocks in the country are part of the crystalline basement complex that underlies a significant portion of West Africa. The basement complex in Nigeria consists of several rock types, including ancient igneous and metamorphic rocks example: granite, gneiss, amphibolite, schist, migmatite these are just a few examples of the basement rocks in Nigeria. Hawal Massif lies within the geographical coordinates, northing's $9^{\circ} 45'$ and $9^{\circ} 51'$ and easting's $12^{\circ} 15'$ and $12^{\circ} 25'$ and covers an area of approximately $23.674.88\text{km}^2$. Hawal massif is bounded in the west and south by the Gongola and Yola arms of the Benue Trough. Bassey et al. (2012) presented a brief account of the geology of the Hawal Basement Complex. The earliest formed crustal materials observed in the area are mafic xenoliths or amphibolites fragments in gneisses and migmatites of probably Proterozoic age (Dada, 1998). The gneisses and migmatites are the most widespread rocks and often occupy the low elevation areas. NW-SE folds, foliation, shear zones and faults mark these metamorphic rocks. Intrusive to them are granitoids of the Pan African orogeny (700 150 Ma). The Pan African deformational direction is mainly NE-SW and N-S. The Pan African deformation affected already existing metamorphic rocks producing polyphase deformations e.g. pre-existing folds were refolded along NE-SW and N-S directions in places producing axial planar foliation. The Cretaceous witnessed the formation of the sedimentary basins, which are infilled with clastic and argillaceous sediments. Mesozoic and Cenozoic magmatism saw the emplacement of basaltic bodies, with the main outcrops at Biu, Song and Michika areas. The study area covers an area of 30.8km^2 on Longitude $12^{\circ}35'00''\text{E}$ - $12^{\circ}38'00''\text{E}$ and Latitude $9^{\circ}48'00''\text{N}$ - $9^{\circ}51'00''\text{N}$.

1.1 STATEMENTS OF PROBLEM

Most of the work done around the area is on regional scale. People in and around the Adamawa Basement have carried out previous work on geology and geochemical analysis. As such, there is need for this project work in order to fill the gap created by the scarcity of recent research in the specific study area of Song (Bongwo Hills) or to complement previous studies carried out around the area.

1.2 AIM OF THE STUDY

The main aim of this work is to carry out a detail geological mapping of Bongwo hill (Three sister's rock) in Song, and to provide petrographic and geochemical studies of various rock samples in the area in order to understand the assemblages of the rocks.

1.3 OBJECTIVES OF THE STUDY

1. To produce a detailed geologic map of the study area.
2. To indicate the different rock types and texture of the rocks found within the study area.
3. To delineate structural features found on the rocks within the study area.
4. To collect rock samples for petrographic analysis to reveal the presence Granite-Gneiss rocks of Bongwo Hill (*Three sister's rocks*).
5. To collect samples for geochemical analysis to reveal the chemical composition of the Granite-Gneiss in Bongwo Hill (*Three sister's fvvffcc rocks*) using Atomic Absorption Spectrophotometry (AAS).

1.4 LOCATION AND ACCESSIBILITY

The study area (Song and its environs) lies between longitudes 12°35'E and 12°38'E, and latitudes 9°48'N and 9°50'N (Fig.1). The Three Sisters Hills are located along the Song-Gombi Road behind the Local Government Secretariat. Near to the Three Sisters' Hills in Song town are: a functional daily market, motor park, health clinic, secondary and primary schools etc. Several footpaths and cattle routes traverse the area. Song is the gateway to the northern cities in Adamawa State.

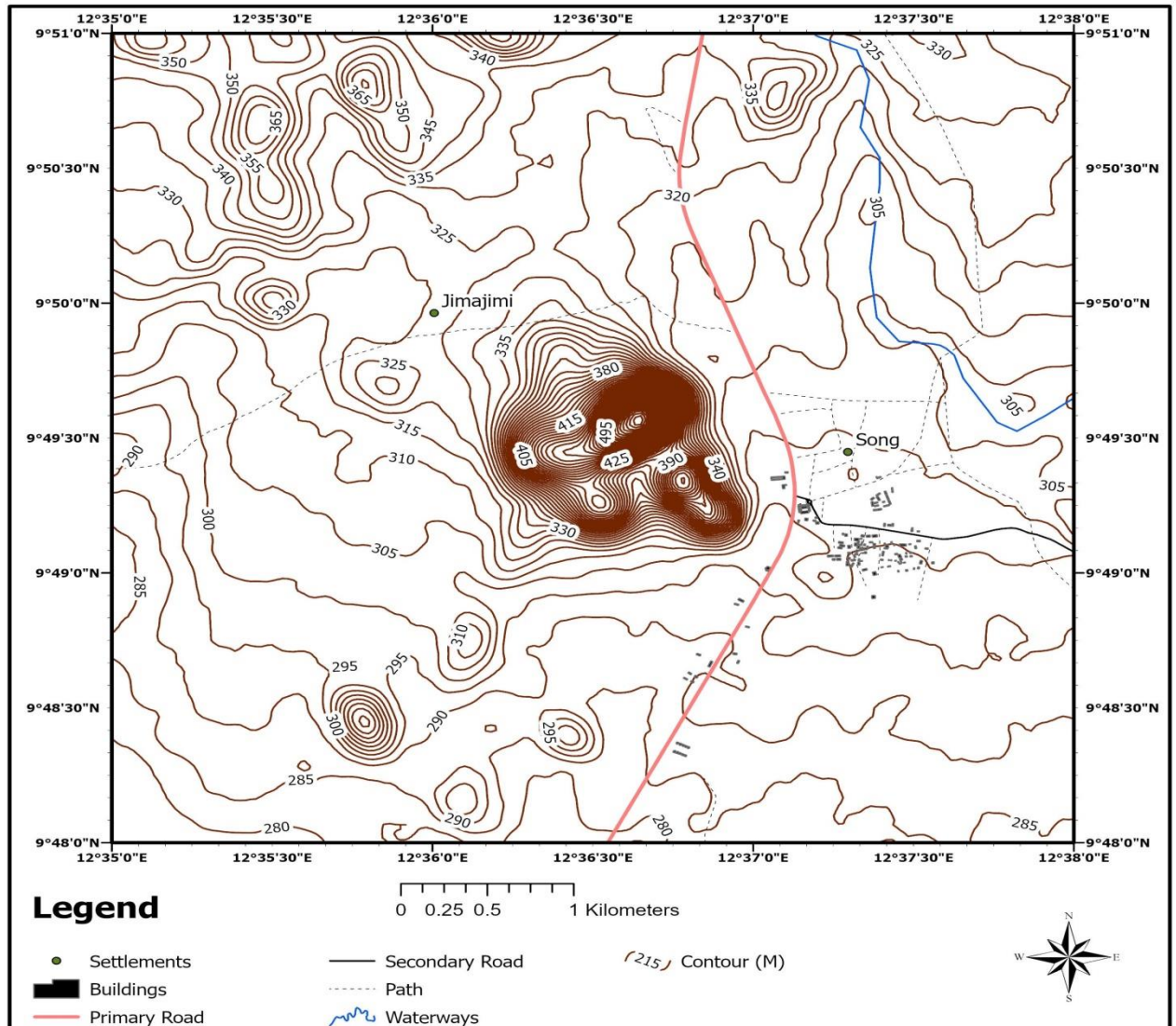


Fig 1.1 Topographic map of a study area

1.5 RELIEF AND DRAINAGE

The Three Sisters' Hill occurs isolated but is part of a group of hills located at the southern part of Hawal basement in NE Nigeria. Some of the hills occur as ridges like at the NE of the study area (Vinde Sawim hills) and at Agurvula in the NW. Other hills are found at Mudore and Tapare. The lowest elevation ranges from 300m to 450m, while the highest peak lies at about 500m above mean sea level. The elevation of the land surface generally increases northwards, with several of the hills marked by steep straight slopes caused by exfoliation and/or tectonism.

The area is well drained by a network of streams whose courses are obviously controlled largely by geologic structures. Most streams have straight channels and flow N-S NW-SE or NE-SW. The major streams in the area have their origin northward and outside of the study area. Where there are many hills. They then flow southwards along narrow channels. River Song follows NNW-SSE courses and flows through the central part of the area. This river and others to the east and west of it form essential parallel to sub parallel drainage configuration.

1.6 CLIMATE

Like in Middle, belt region of Nigeria two main climatic seasons are experienced in the area, each lasting about six-months duration the dry hot (with a short duration low temperature or hamathan period) and dusty season commences from November to April. The rainy season with moderate to high intensity precipitation commences from April and lasts till October. The peak period of the rainy is July – September.

1.7 SCOPE OF THE WORK

The scope of this is to study the geology of part of Song LGA of Adamawa by carrying out a detail geological field mapping and carrying out petrographic and geochemical analysis of selected rock samples.

CHAPTER TWO

LITERATURE REVIEW

2.1 REGIONAL GEOLOGY

Nigeria lies east of the West African Craton and northwest of the Congo Craton, within a region significantly influenced by the Pan-African orogeny (600 ± 150 Ma). The Nigerian Basement Complex, part of the Trans-Saharan Belt, lies south of the Tuareg Shield and was formed through the collision between the passive margin of the West African Craton and the active Pharusian Belt of the Tuareg Shield around 600 Ma (Burke and Dewey, 1972; Caby et al., 1981). This collision reactivated the internal regions of the belt, leading to the deformation, metamorphism, and emplacement of granitic rocks in the Nigerian Basement Complex (Rahman, 1988; Dada, 2006). The Basement Complex comprises migmatites, gneisses, schist's, and older granites, reflecting multiple phases of deformation and metamorphism during the Pan-African event (Grant, 1978; Odeyemi, 1988). Earlier tectonic events, such as the Liberian (2.7 Ga) and Eburnean (2.0 Ga) orogenies, also contributed to the formation of Archean and Paleoproterozoic crustal components within the complex (Dada, 2006; Fitches et al., 1985). Overlying the Precambrian basement are younger sedimentary basins, such as the Benue Trough and Niger Delta Basin, which formed during the Mesozoic and Cenozoic due to rifting associated with the opening of the South Atlantic Ocean (Benkhelil, 1989; Guiraud et al., 1992). These basins are economically significant due to their hydrocarbon resources.

2.2 LITHOLOGICAL UNITS OF THE NIGERIAN BASEMENT TERRAIN

The Nigerian Basement Complex is divided into three broad lithological groups: Precambrian Basement Complex which composed mainly of migmatite gneiss, Older Granites, and metasediments (schist belts), Triassic-Jurassic Younger Granite Province with an anorogenic granite suite located in north-central Nigeria, and Sedimentary Basins That include the Chad Basin (northeast), Sokoto Basin (northwest), Niger Delta (south), Benue Trough (linking the Chad Basin to the Niger Delta), and the Niger (Nupe) Basin (forming an aborted link between the Sokoto Basin and the Niger Delta) (Rahman, 1989).

The Nigerian Basement Complex forms the southern part of the Trans-Saharan Mobile Belt, a Neoproterozoic (750–500 Ma) structure situated between the Archean-Paleoproterozoic West African Craton (west), the Congo Craton (southeast), and the East Saharan Block (northeast) (Caby, 1987; Black, 1980). It is part of the Pan-African Mobile Belt, lying between the West African and Congo Cratons and south of the Tuareg Shield. The complex was significantly

affected by the Pan-African orogeny (600 ± 150 Ma), which resulted from the collision between the passive continental margin of the West African Craton and the active Pharusian Continental margin (Burke and Dewey, 1972; Dada, 2006). This event led to intense deformation, regional metamorphism, migmatization, and granitization, producing syntectonic granites and homogeneous gneisses (Abaa, 1983). The Pan-African orogeny was the culmination of three major orogenic cycles: Liberian Cycle ($2,700 \pm 200$ Ma), Eburnean Cycle ($2,000 \pm 200$ Ma) and Pan-African Cycle (600 ± 150 Ma). These cycles were characterized by intense deformation, isoclinal folding, and regional metamorphism, followed by migmatization and granitization. The late stages of the Pan-African orogeny involved the emplacement of granites and granodiorites, accompanied by contact metamorphism, faulting, and fracturing (Gandu et al., 1986; Olayinka, 1992).

Studies by McCurry (1971) and others have highlighted the structural and metamorphic evolution of the Basement Complex. McCurry identified two phases of intense deformation in northern Nigeria, resulting in tight isoclinal folding with E-NE–W-SW and N-S trends. She concluded that the Pan-African event reached orogenic status in northwestern Nigeria, with the crystalline basement reactivated and the supracrustal cover regionally metamorphosed.

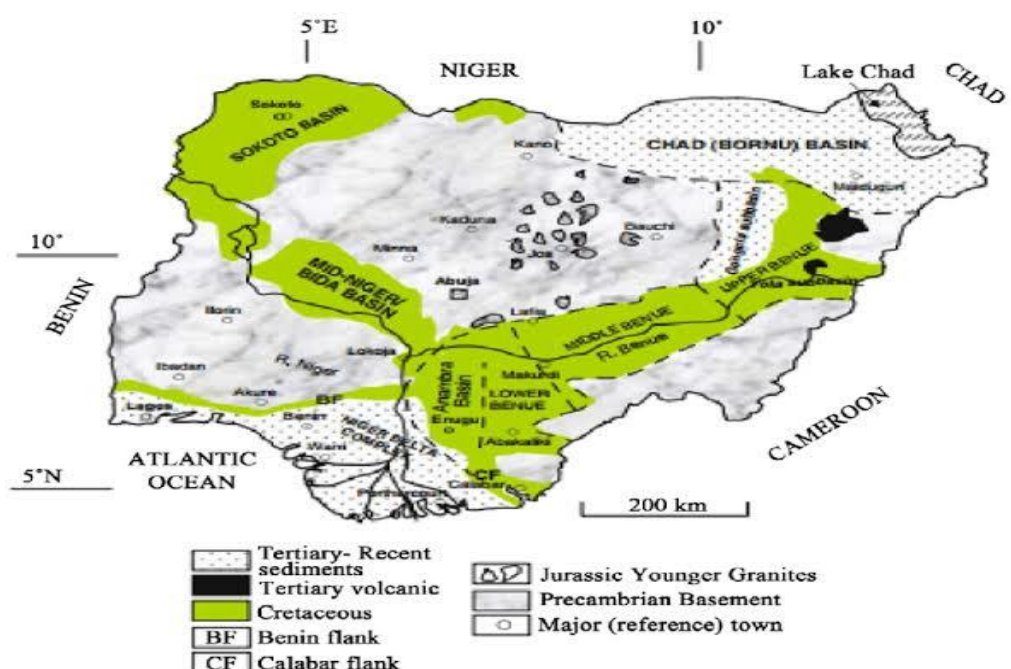


Fig 2.1 Regional Geology of Nigeria

2.3 REVIEW OF PREVIOUS WORKS

Most of the previous works in the study area were regional in extent. These include;

Yaro *et al* (2018) studied the part of the Hawal Basement Complex with the aim of identifying volcanic basic rocks in the area to see the possibility of occurrence of economic mineral deposits in the study area.

Bassey, (2007) studied the phenomenal mass waste in Song area of the Hawal Basement, NE Nigeria the study revealed that the mass wasting is caused by a combination of lithological, structural, topographic, climatic and organic factors.

Obiefuna and Ibrahim, (2014) studied the mineralogical composition of Biu Basalt deposit, northeastern Nigeria. From their studies, they discovered that the basaltic rocks of the study area are tholeiitic basalt and they show little petrographic variations which is related to their mode of occurrence and origin.

Kwache and Ntekim, (2015) studied The Geology of Dumne Area in Southeastern Hawal Massif, Northeastern Nigeria. In their words they reported that Granite gneiss constitute the predominant rocks types in Dumne area of the southern Hawal massif. Other major rocks include granites, migmatites, older granites and basalts. The rocks are granitic in composition and exhibit close relationship mineralogically, geochemically and in the field. The basalts are basinites.

Ekwueme (1993) gave some details on the basement rocks of the North eastern Nigeria in which He classified the rocks to consist of Migmatites, Gneisses, Diorites, Porphyritic granites, Volcanic rocks and Alluvial deposits.

Saleh and Maunde, (2017) studied the geology and petrography of the Uddin granitic Hills and environs, part of the Hawal massif Northeastern Nigeria. From Geologically, Uddin area mainly comprises of crystalline basement rocks, geomorphologically represented by both low and high lands. The gneisses mostly cover the low lands, while the high lands are exclusively granitic. Metamorphism is restricted to the lowland areas and it extends gradationally to the relatively higher lands. Migmatites part of the basement is conspicuously absent, probably hidden underneath the exposed rocks. The granites in the area are exclusively large grain sized characterized by sets of phenocrysts.

CHAPTER THREE

MATERIALS AND METHOD

3.1 MATERIALS

The field equipment used for the mapping are:

- Global Positioning System (GPS)
- Compass clinometers
- Hammer and chisel
- Hand lens
- Measuring tape
- Study area topographic map
- Masking tape
- Camera
- Field notebook, pencil and biro.

3.2 METHODS

This research work involved mainly the geologic description and geochemical study of Bongwo hills (Three sister's rock) in Song, Northeast Nigeria. This research comprised Four (4) methods.

1. Reconnaissance survey
2. Field mapping and sampling
3. Petrographic studies
4. Geochemical analysis

3.2.1 RECONNAISSANCE SURVEY

In this method of field study, I was able to ascertain the accuracy of the base map where I justified what i saw on the topographic map and make some correction where necessary, such as presence of new settlements, weathered rock materials, stream channels, farmlands etc.

3.2.2 GEOLOGIC MAPPING

Geological field mapping of the study area was carried out using the above-mentioned materials and identified the geological formation present and data obtained was used to created geological map, I was able to determined structures and mineral potentials of the rocks. Samples were collected from the study area and sealed in polythene bags for petrographic and geochemical analysis.

There are three (3) types of mapping techniques- the traverse, contact and exposure mapping methods (Greenly and Williams, 1930; Haruna, 2022). For the purpose of this study exposure technique was adopted as its best suites the aim and objectives of the study.

3.2.3 PETROGRAPHIC ANALYSIS

This involved examining thin sections of the rock samples collected under a microscope, I identified the minerals and textures visible in the rock. Each piece was polished to a 30-micron thick slide for study using the petrographic microscope attached with a camera. The slides were observed under plane polarized (PPL) and crossed polarized light (XPL).

3.2.4 GEOCHEMICAL ANALYSIS

Geochemical analysis was carried out to determine the concentration of major elements present in the rock sample. Representative samples were crushed, powdered, and digested using suitable acid solution. The resulting solutions were analyzed using Atomic Absorption Spectrophotometry (AAS). This method measure the amount of each element based on absorption of light by atoms. Major elements such as Si, Fe, Ca, Mg, Na, K, and Mn were determined and the results were used to classify the rock and interpret their geochemical characteristics.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.0 RESULTS PRESENTATION

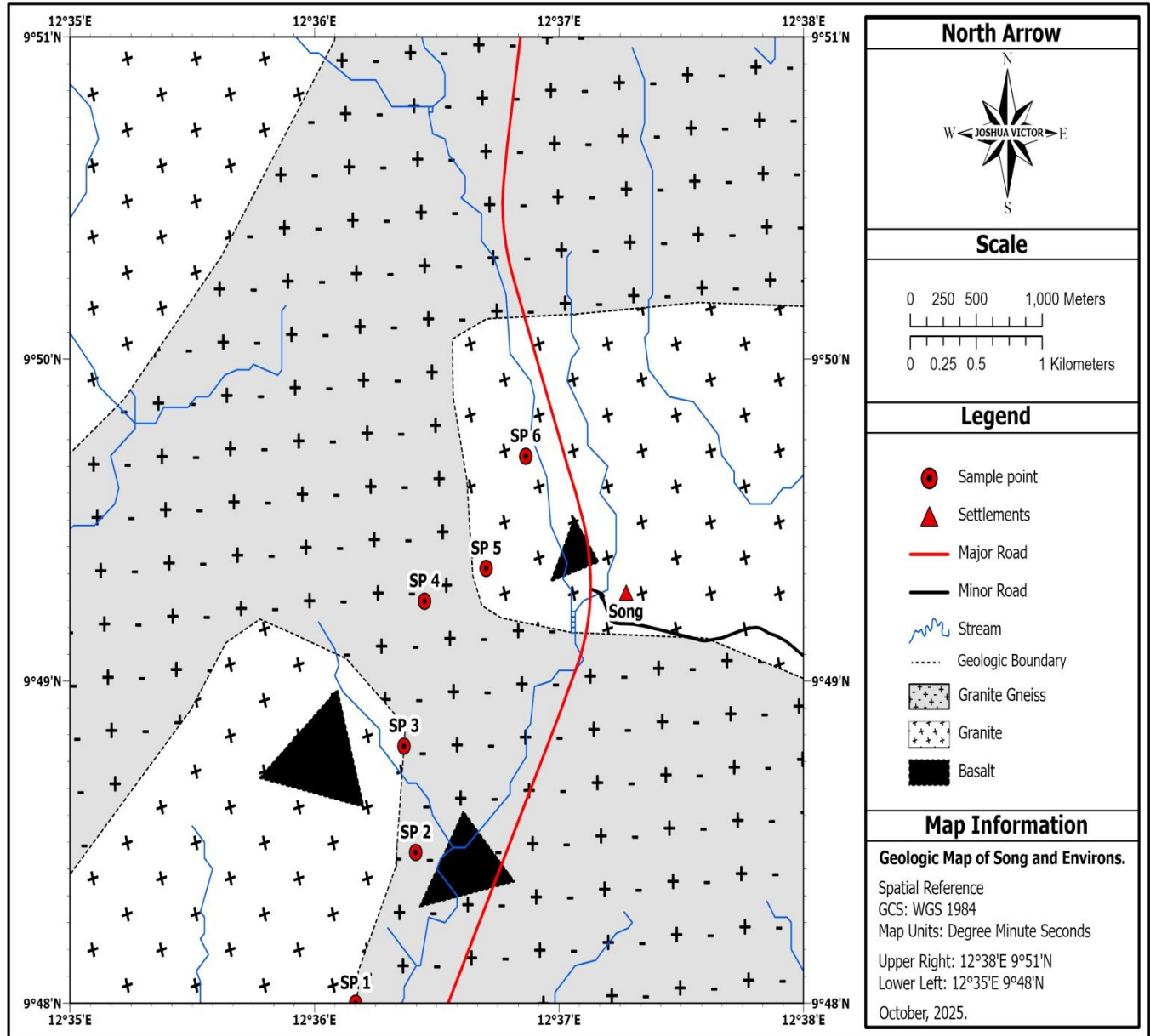


Fig 4.1: Geological Map of the study area

4.1 LITHOLOGY

The study area is underlain predominantly by Basement Complex rocks of Precambrian age. The main lithological units mapped include phaneritic granite, coarse to medium-grained granite, gneiss, migmatite, basalt, and pumice. These rocks occur as discrete outcrops and in places show gradational or sharp intrusive contacts.

4.1.1 PHANERITIC GRANITE

In hand specimen, the granite is light grey to pink, holocrystalline and coarse grained. The dominant minerals include quartz, K-feldspar, plagioclase and biotite. The granite occurs as massive outcrops, boulder ridges, and isolated domes. Weathering has produced spheroidal boulders scattered across the landscape.



Fig 4.2 Showing Phaneritic granite in the field

4.1.2 COARSE TO MEDIUM GRAINED GRANITE

This unit is generally light coloured with a phaneritic texture. It is composed of quartz, orthoclase, plagioclase, and biotite, with occasional muscovite. Some exposures show pegmatitic veins. Textural variation was observed within the granite body, with fine to medium-grained margins grading into coarse interiors.



Fig 4.3 Showing Coarse-Medium grain granite in the field

4.1.3 GNEISS

The gneiss in the study area is banded, showing alternating light coloured quartzo–feldspathic layers and dark mafic bands rich in biotite and hornblende. The foliation is well developed and trends mainly NE–SW. Gneissic outcrops occur at the base of the Three Sisters Hills and are often associated with shear zones.



Fig 4.4 Gneiss showing banding of light and dark minerals in the field

4.1.4 MIGMATITE

Migmatite outcrops occur near the contact between granite and gneiss. They show evidence of partial melting with streaks of leucosome and melanosome. Stromatic and nebulitic structures are visible, indicating high-grade metamorphic conditions.



Fig 4.5 Migmatite showing partial melting

4.1.5 BASALT

Basalt occurs as laterally continuous flows and sills intruding the basement rocks. In hand specimen, the basalt is dark grey to black, fine grained, and massive. Some exposures show columnar jointing.



Fig 4.6 Basalt intruding the basement rock in the field

4.1.6 PUMICE

A small deposit of pumice was discovered near a stream close to the major rock exposures. The pumice is vesicular, light in weight, and dark in color. Its occurrence indicates explosive volcanic activity in the geological past.



Fig 4.7 Showing Pumice near stream close to the three sister's rock

4.2 PETROGRAPHY

Five (5) thin sections from the different representative samples were studied in detail and the rocks identified. The samples are S1, S2, S3, S4, and S6. The rocks studied in the area include Phaneritic granite, Coarse-Medium grain biotite granite.

Five (5) different samples were selected for the petrographic analysis of the study area (S1, S2, S3, S4, and S6). Data from the study area and all rock samples from the project are presented in tables, maps, plots and photographs (plates).

Table 4.1: Sample points, their coordinates and rock types

S\N	Sample points	Rock type	Northing	Easting	Elevation
1	S1	Phaneritic Granite	9° 48'20"	12°36'10"	298m
2	S2	Coarse Grain Granite	9° 48'28"	12°36'25"	283m
3	S3	Coarse Granite	9° 49'15"	12°36'27"	309m
4	S4	Medium Grain Biotite Granite	9° 49'21"	12°36'42"	310m
5	S6	Medium Grain Granite	9° 49'42"	12°36' 52"	304m

4.2.1 Megascopic Description:

The granite samples are phaneritic, holocrystalline, and equigranular, with medium- to coarse-grained crystals that are clearly visible to the naked eye. The rocks are light grey to pinkish, dominated by quartz, feldspar (orthoclase, microcline, plagioclase), and biotite.

4.2.2 General Microscopic Description of Samples:

Most of these minerals seem to have been affected by slight weathering as they show minor alterations on their surface. Biotite in the rock is identified as black specks and can easily be seen with the naked eye. In hand specimen the rock is light to black coloured, holocrystalline and coarse grained. Large phenocrysts of quartz and feldspar are clearly visible in the rock.

4.2.3 Phaneritic Granite (Sample S1)

In thin section, the phaneritic granite consists mainly of quartz, feldspars (plagioclase, orthoclase and microcline), biotite with subordinate muscovite and opaque. Quartz occurs as anhedral to subhedral grains; it is colorless under plane polarized light and appears white in crossed polars. It has low relief, low birefringence and shows undulose extinction. The plagioclase is subhedral in shape, colorless in plane polarized light and milky to grey in crossed polars. It displays albite twinning and zoning due to deformation, with minor

alteration to sericite. Orthoclase occurs as cloudy subhedral crystals showing simple twinning, while its characteristic crosshatched twinning identifies microcline. Biotite appears as brown pleochroic flakes with perfect basal cleavage, it has moderate relief and sometimes shows alteration along cleavage planes to chlorite and opaque minerals. Muscovite occurs as colorless flakes with basal cleavage and weak birefringence, locally replacing feldspars. Opaque are scattered as small dark grains, isotropic under crossed polars. The texture is holocrystalline and equigranular, with all the minerals nearly uniform in size, confirming the phaneritic nature of the rock.

4.2.4 Microscopic Description of Phaneritic Granite

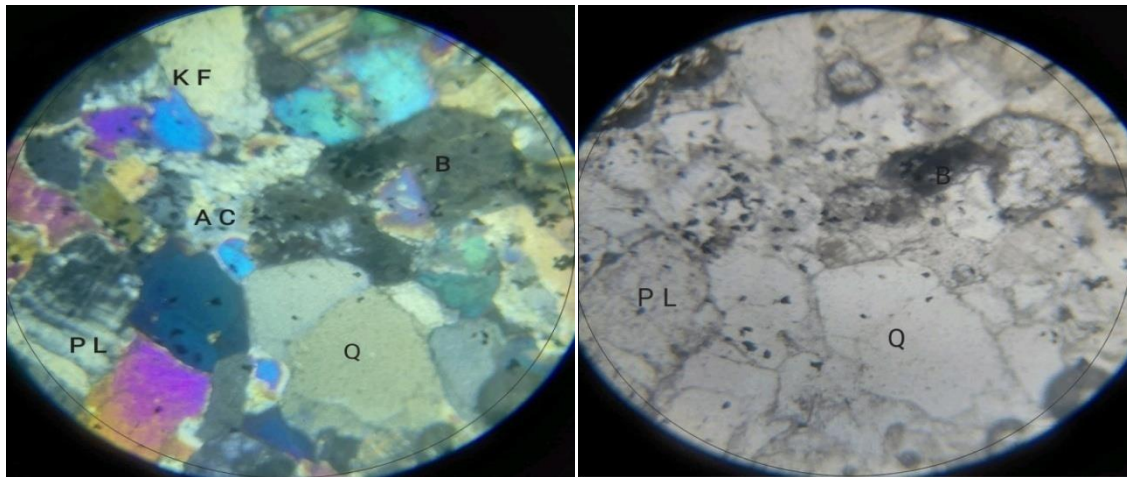


Plate I: Photomicrograph of Phaneritic granite (sample S1), under (a) Crossed polarized light (XPL) (b) Plane polarized light (PPL).

SHOWING

(B)=Biotite

(Q)=Quartz,

(PL) =Plagioclase

(KF) =K-Feldspar

(AC) =Accessory minerals

Mag. =X100

Modal composition or estimate of minerals in (S1)

Minerals	Composition (%)
Quartz	38
Plagioclase.....	27
K-Feldspar	19
Biotite.....	13
Accessory minerals.....	3

4.2.5 Coarse Grain Granite (Sample S2)

The hand specimen of the coarse-grained granite displays a light grey to whitish color, with visible large grains of quartz, feldspar, and dark mica minerals. The rock is holocrystalline, phaneritic, and coarse-grained, with individual crystals generally >5 mm in size. Quartz appears as glassy, translucent grains, feldspar as whitish to pinkish crystals, and biotite as dark brown to black flaky minerals. No distinct foliation or compositional banding is visible macroscopically, confirming its igneous origin.

4.2.6 Microscopic Description of Coarse Grain Granite

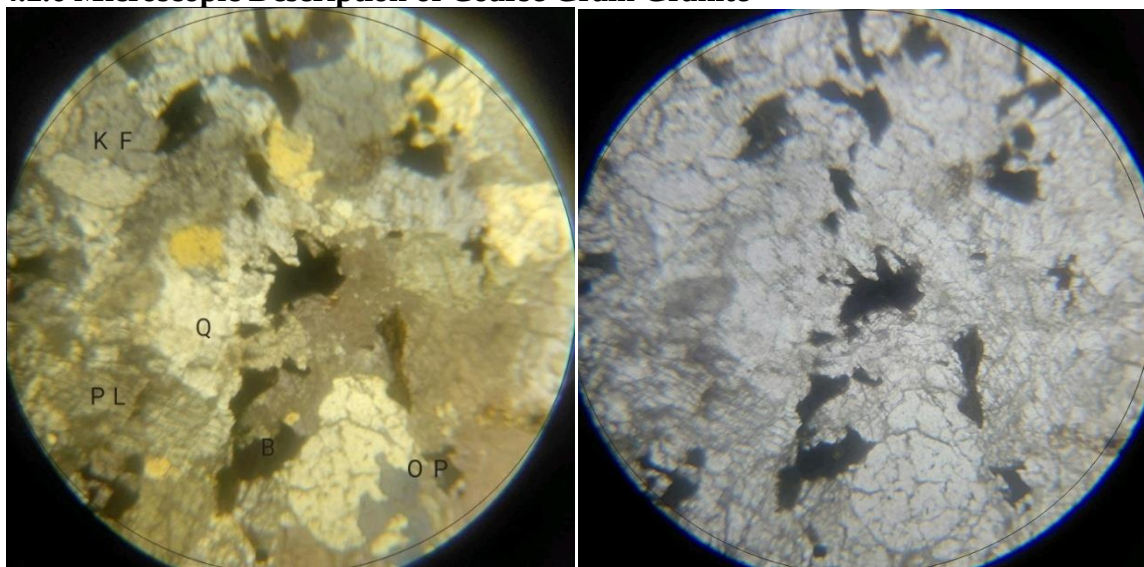


Plate II: Photomicrograph of Coarse Grain Granite (Sample S2), under (a) Crossed polarized light (XPL) (b) Plane polarized light (PPL).

SHOWING

(B)=Biotite

(Q)=Quartz

(PL) =Plagioclase

(KF) =K-Feldspar

(OP)=Opaque

Mag. =X100.

Modal composition or estimate of minerals in (S2)

Minerals	Composition (%)
Quartz.....	40
Plagioclase	30
K-Feldspar.....	20
Biotite.....	7
Opagues.....	3

4.2.7 Coarse Granite (Sample S3)

This sample represents coarse-grained granite, a typical plutonic igneous rock formed from the slow crystallization of silica-rich magma in the continental crust. The rock exhibits a holocrystalline texture, characterized by interlocking, anhedral to subhedral grains of varying sizes, and indicative of equigranular texture common in granites. No significant foliation or deformation features are evident, suggesting a relatively undeformed history post-crystallization. An intergrowth of quartz in plagioclase is observed at some feldspar-quartz boundaries, likely resulting from metasomatic alteration, which imparts a wormy appearance to affected grain edges.

4.2.8 Microscopic Description of Coarse Granite

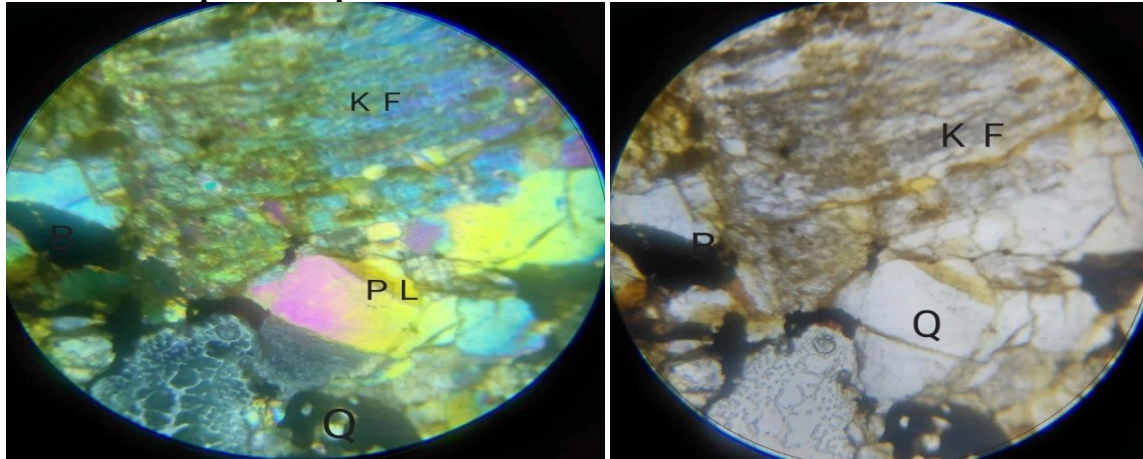


Plate III: Photomicrograph of Coarse Granite (Sample S3), under (a) Crossed polarized light (XPL) (b) Plane polarized light (PPL).

SHOWING

(B)=Biotite

(Q)=Quartz

(PL) =Plagioclase

(KF) =K-Feldspar

(OP)=Opaque

Mag. =X40.

Modal composition or estimate of minerals in (S3)

Minerals	Composition (%)
Quartz.....	27
Plagioclase	25
K-Feldspar.....	38
Biotite.....	8
Accessories'	2

4.2.9 Medium Grained Biotite Granite (Sample S4)

The sample is a holocrystalline, medium-grained biotite granite composed mainly of quartz, plagioclase, and K-feldspar, with subordinate biotite and minor opaque/accessories. The rock shows an interlocking, equigranular texture, with quartz displaying undulose extinction,

plagioclase showing polysynthetic twinning, K-feldspar locally perthitic, and biotite occurring as pleochroic flakes. Weak strain effects are present but no strong foliation is observed.

4.2.10 Microscopic Description of Medium Grain Granite

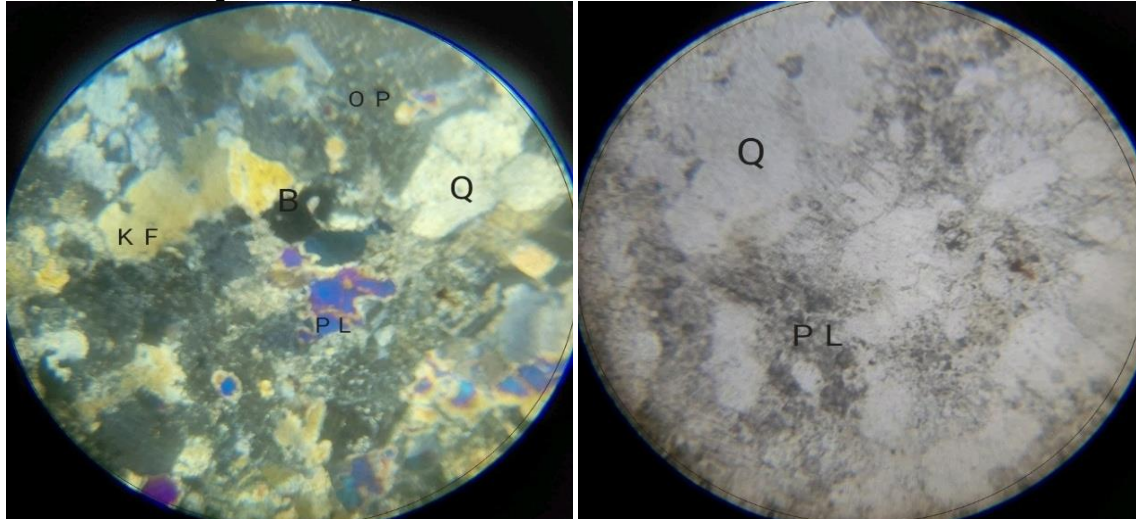


Plate IV: Photomicrograph of Medium Grain Granite (Sample S4), under (a) Crossed polarized light (XPL) (b) Plane polarized light (PPL).

SHOWING

(B)=Biotite

(Q)=Quartz

(PL) =Plagioclase

(KF) =K-Feldspar

(OP)=Opaque

Mag. =X100.

Modal composition or estimate of minerals in (S4)

Minerals	Composition (%)
Quartz.....	35
Plagioclase	30
K-Feldspar.....	25
Biotite.....	8
Accessories'	2

4.2.11 Medium Grained Granite (Sample S6)

The thin section of the medium-grained granite reveals a holocrystalline and interlocking texture. Quartz occurs as anhedral grains showing undulose extinction, while plagioclase displays polysynthetic twinning with minor alteration. K-feldspar is present as subhedral crystals, locally showing perthitic texture, and biotite appears as pleochroic brown flakes with partial alteration to chlorite. The overall texture is equigranular, and the mineral assemblage reflects crystallization from a silica-rich melt at shallow crustal levels.

4.2.12 Microscopic Description of Medium Granite

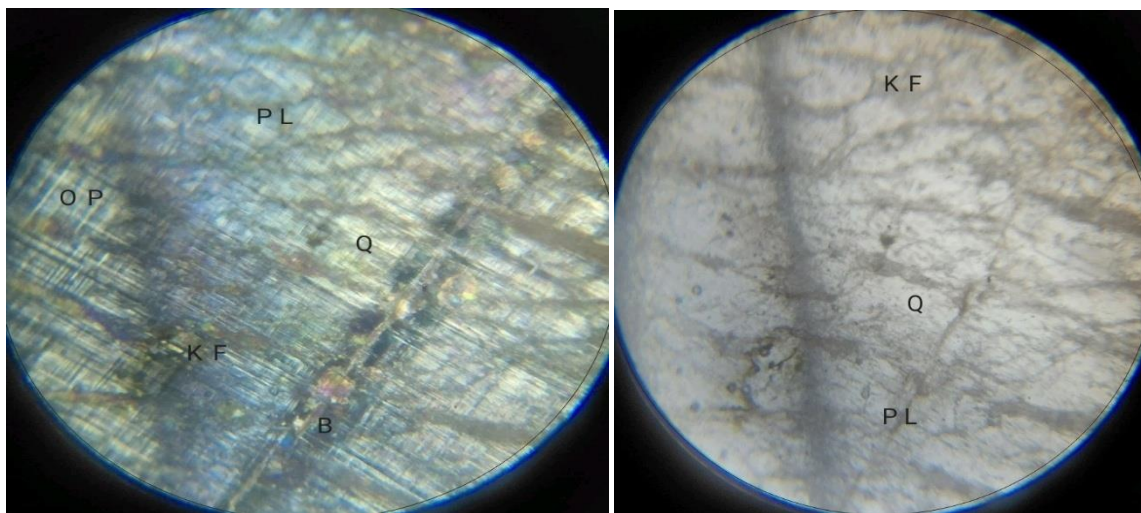


Plate V: Photomicrograph of Medium Granite, (sample S6) under (a) Cross polarized light (XPL) (b) Plane polarized light (PPL).

SHOWING

(B)=Biotite

(Q)=Quartz

(PL) =Plagioclase

(KF) =K-Feldspar

(OP)=Opaque

Mag. =X100.

Modal composition or estimate of minerals in (S6)

Minerals	Composition (%)
Plagioclase	44
Quartz.....	35
K-Feldspar.....	11
Biotite.....	7
Opaques.....	3

4.3 GEOLOGICAL STRUCTURES

Field mapping revealed several structural features affecting the rocks in the study area. These include joints, fractures, foliations, and lineations.

4.3.1 Joints

Joints are the most common structures in the granites and basalts. They occur as systematic sets trending predominantly NE–SW and NW–SE. Some exposures show columnar joints in basalt flows.



Fig 4.8 Showing joint in the field

4.3.2 Fractures

Fractures are widely developed, especially within granites and gneisses. They vary in orientation and density, and are often associated with lateritic fillings. Fractures enhance weathering and erosion of the basement rocks.

4.3.3 Foliation

Well-developed foliation was observed in gneiss and migmatite. It is defined by the alignment of biotite and hornblende within the rocks. The foliation generally trends NE–SW, parallel to the regional Pan-African structural grain.

4.3.4 Lineation

Mineral lineations occur in some gneissic and migmatitic exposures, marked by the alignment of elongated quartz and feldspar grains. Stretching lineation was also observed in sheared gneissic zones.



Fig 4.9 Showing lineation in the field

Table 4.2: Strike and dip of structures in the study area

S/N	LOCATION	STRIKE	DIP
1	N 9° 48' 20" E 12° 36' 10"	017° NE	85°SW
2	N 9° 48' 28" E 12° 36' 25"	028°NE	45°SW
3	N 9° 48' 48" E 12° 36' 22"	042°NW	35°SE
4	N 9° 49' 15" E 12° 36' 27"	053°NE	40°SW
5	N 9° 49' 21" E 12° 36' 42"	085°NW	35°SE
6	N 9° 49' 42" E 12° 36' 52"	092°NE	40°SW

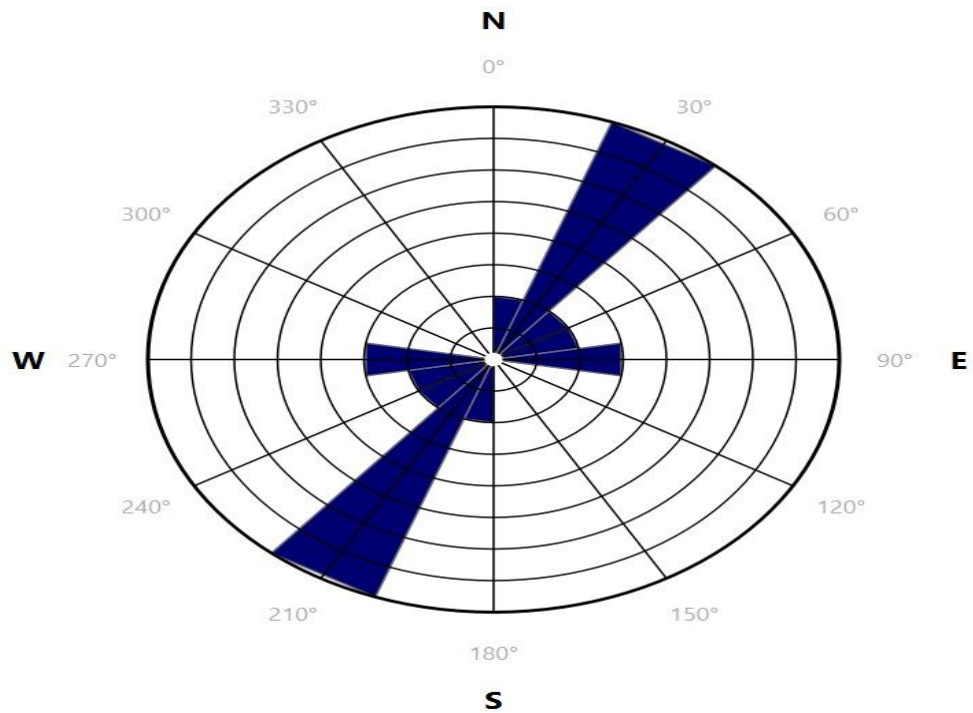


Fig 4.10 Rose diagram showing structural orientations

4.4 GEOCHEMICAL ANALYSIS

The quantitative and qualitative determination of elements in the five (5) representative rock samples was conducted using Atomic Absorption Spectrophotometer (AAS). The analysis was carried out at the Modibbo Adama University, Yola Central Laboratory.

Major elements determined included Silicon (Si), Calcium (Ca), Aluminum (Al), Iron (Fe), Sodium (Na), Potassium (K) and Magnesium (Mg). The elemental components of these elements were determined.

Table 4.3 Shows variation in chemical data with SiO₂ ranging from 65.80-98.70% for the rock samples while Al₂O₃ range from 9.70-36.50%. Ranges of other Oxides are Fe₂O₃ (40.90-64.60%), CaO (6.96-43.50%), Na₂O (28.50-40.20%), K₂O (60.50-198.20%), and MgO (7.20%-13.7%).

S/N	Sample No	Elemental Concentration in Oxide Form							Total Concentration
		SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	Na ₂ O	K ₂ O	MgO	
1	S1	82.40	28.20	42.50	6.96	31.40	70.70	9.0	271.16
2	S2	98.70	14.50	40.90	36.30	40.20	85.30	10.1.	315.9
3	S3	78.10	36.50	51.30	38.70	32.20	73.80	13.7	324.3
4	S4	65.80	9.70	50.20	7.50	28.50	55.50	7.20	224.4
5	S6	98.60	30.10	64.60	43.50	32.90	92.20	10.20	372.1
	Total	424.2	199	249.5	132.96	165.2	377.5	52.2	
	Average	84.84	39.8	49.9	26.592	33.04	75.5	10.04	

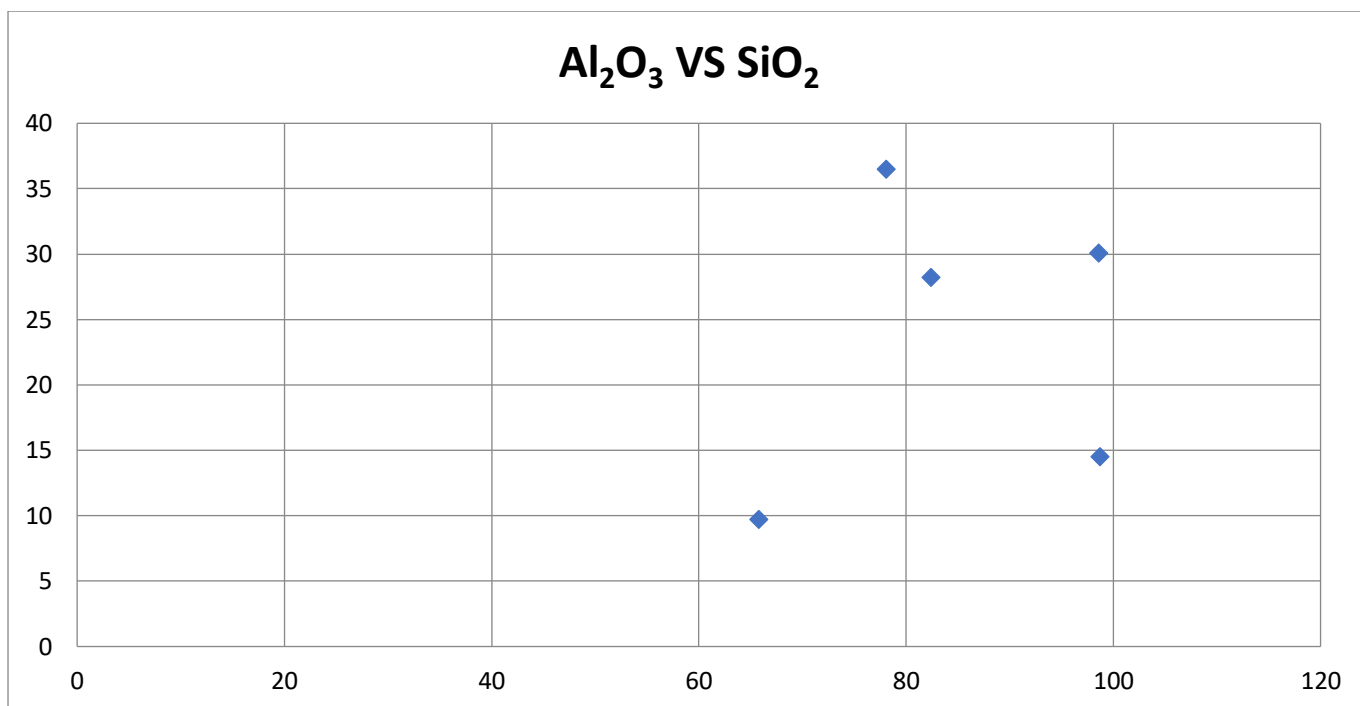


Fig 4.11 Harker variation plot of SiO_2 against Al_2O_3

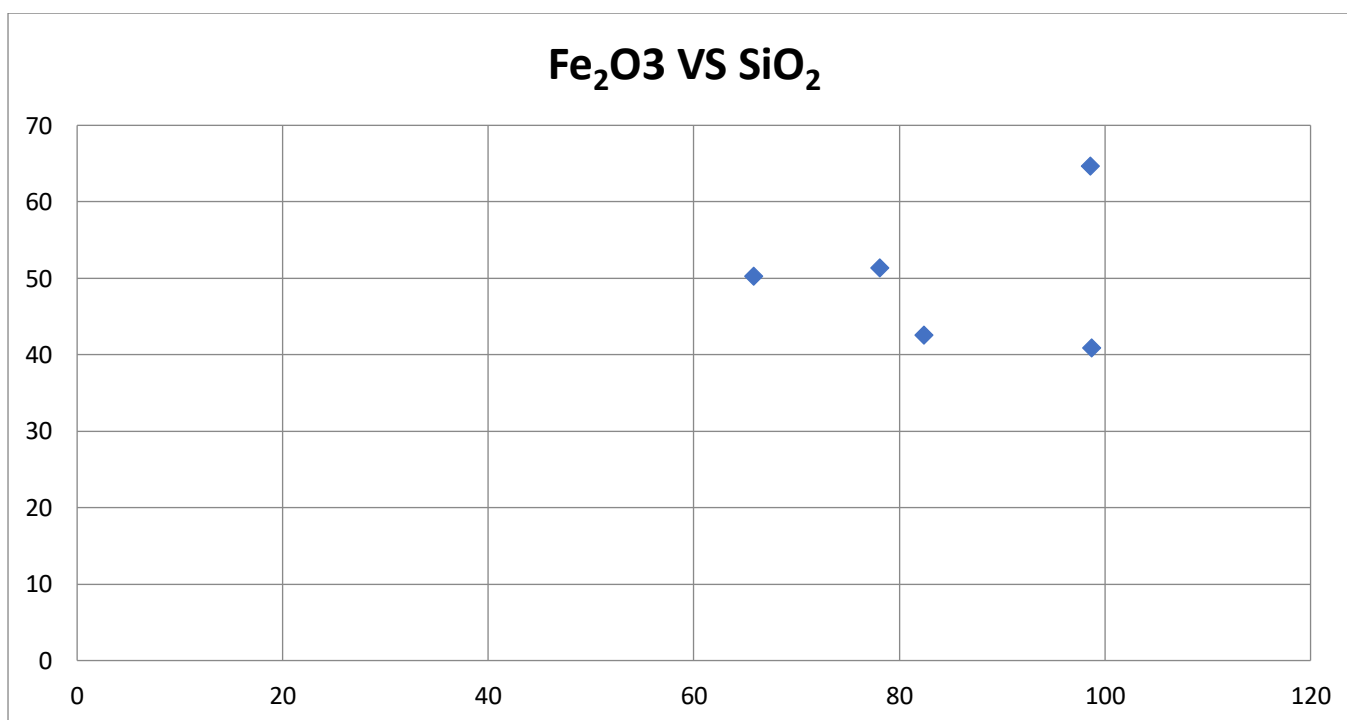


Fig 4.12 Harker variation plot of SiO_2 against Fe_2O_3

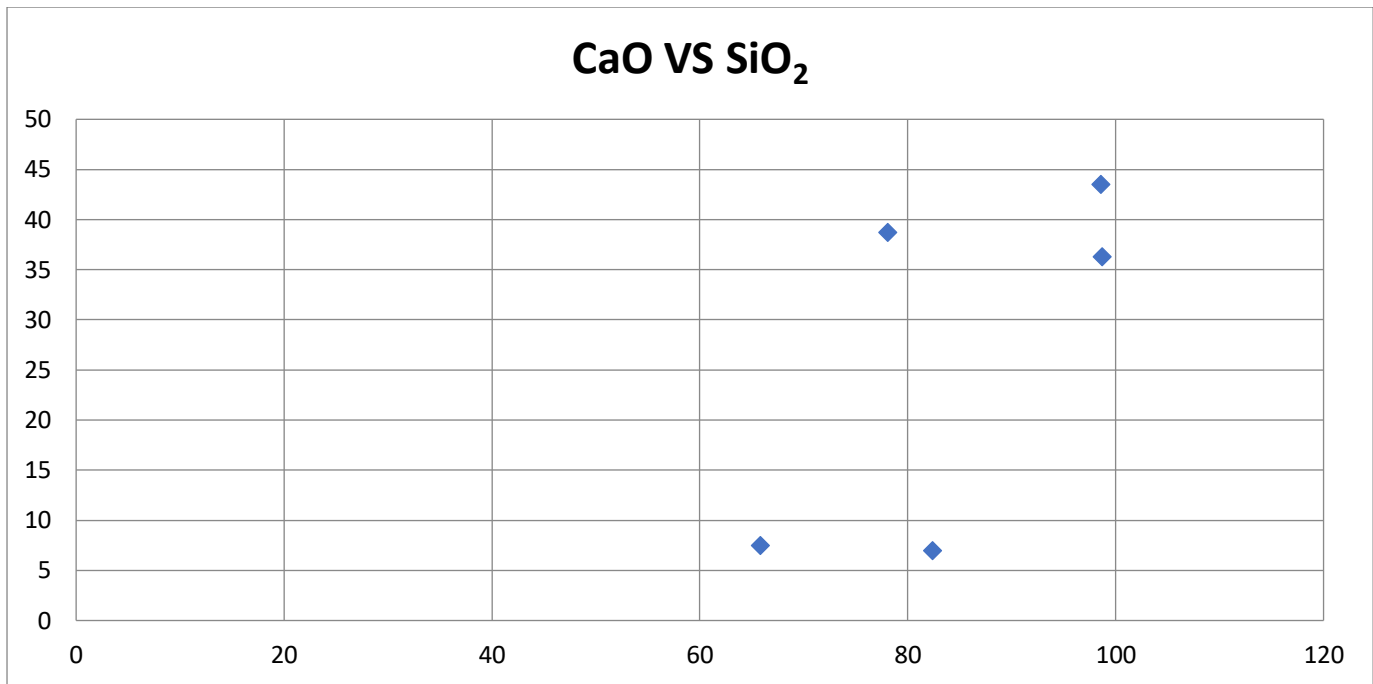


Fig 4.13 Harker variation plot of SiO₂ against CaO

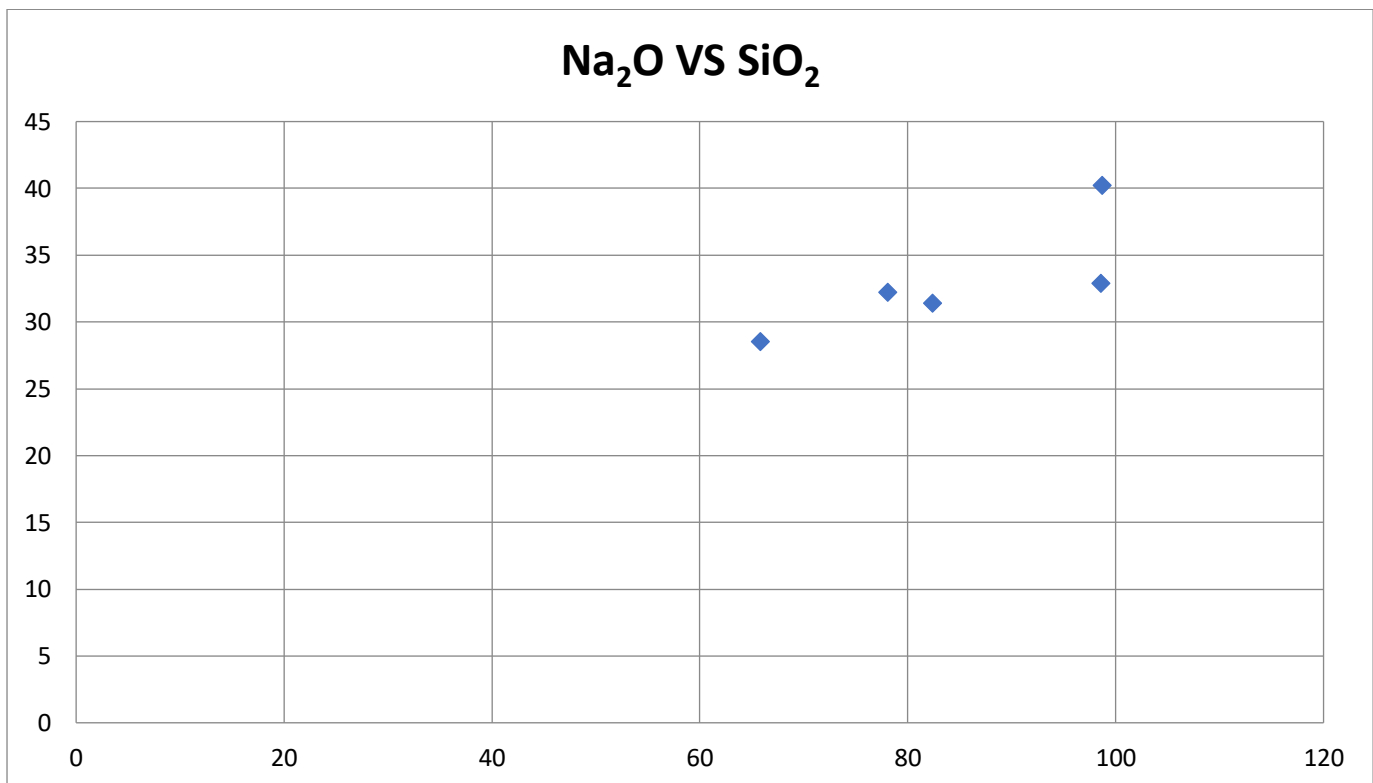


Fig 4.14 Harker variation plot of SiO₂ against Na₂O

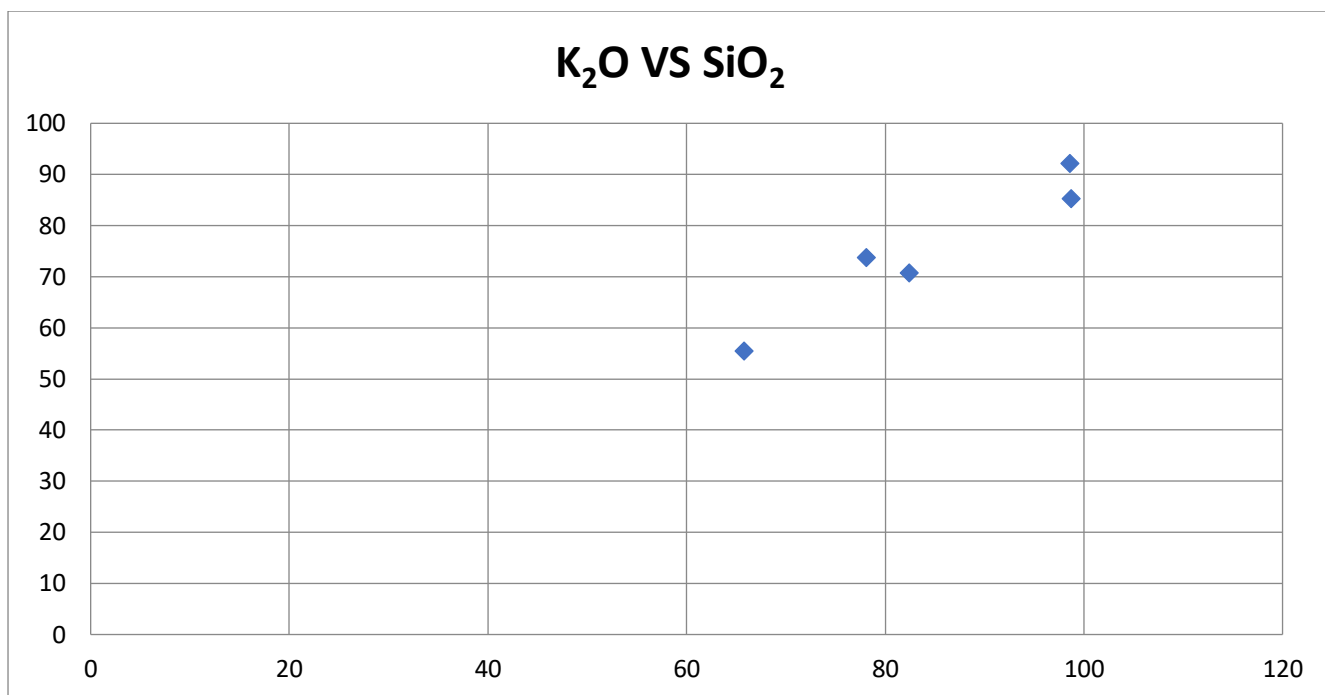


Fig 4.15 Harker variation plot of SiO₂ against K₂O

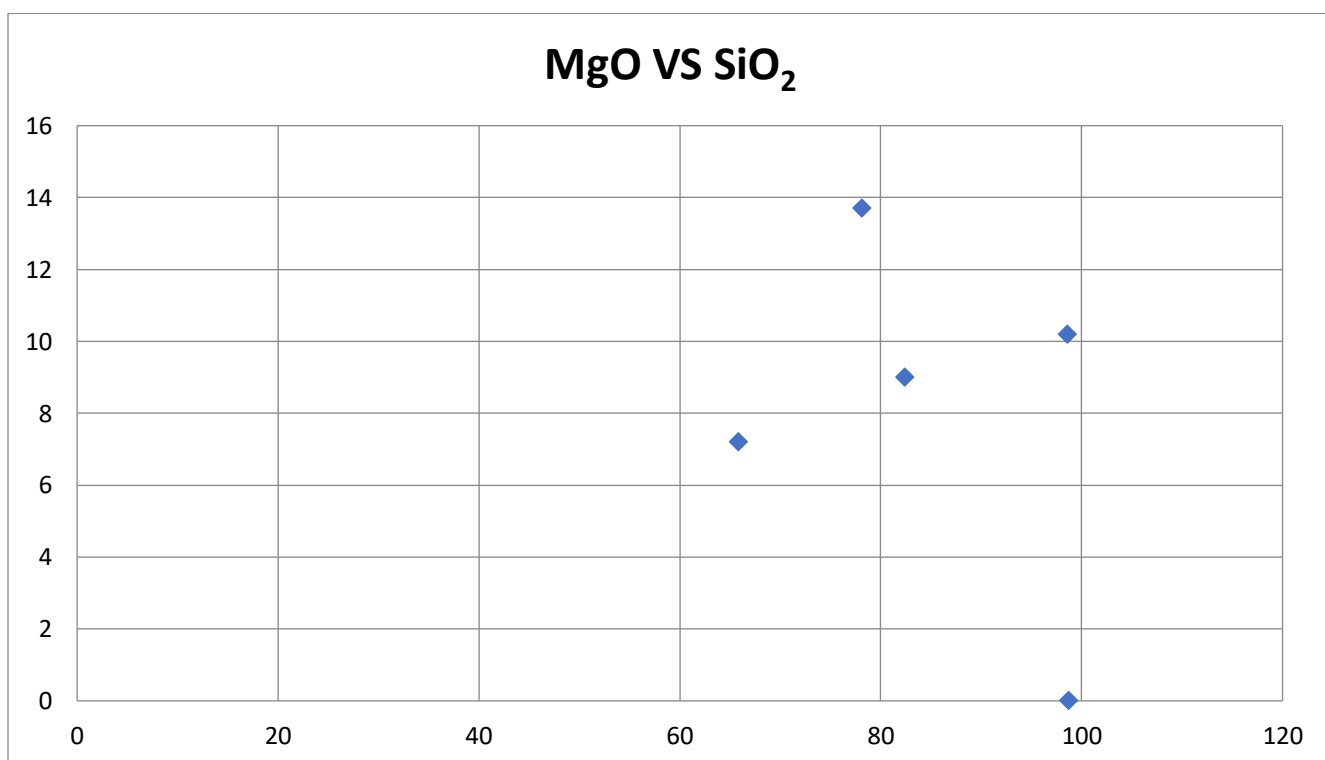


Fig 4.16 Harker variation plot of SiO₂ against MgO

4.4.1 HARKER VARIATION DIAGRAM

The geochemical variations shown by Harker plots (Fig 4.11-4.16) are indicating the relationship between the rocks, which is also an indication of possible fractional crystallization during formation from common source. Most of the rocks correlate with SiO_2 . However, Al_2O_3 , Fe_2O_3 , CaO , MgO and Na_2O have weak correlation as they scattered plots due to possible accumulation of various components and mixture during fractional crystallization. The plot of K_2O against silica has strong correlation indicating silica as the major element during the formation of the rock. The plots indicate that most of the elements show systematic variation with respect to SiO_2 contents.

The field laboratory studies confirm that a Bongwo hill (Three sister's rock) belongs to the Precambrian rocks formed from series of orogenic circle within the mobile belt of central Africa.

4.4.2 GEOCHEMISTRY

4.4.2.1 Silica (SiO_2) Content

All the oxides decrease in their content with increasing silica (Table 4.3). This indicates the removal of the high temperature and plagioclase in the cause of crystallization although the increase in the SiO_2 may indicate more fractionation. Igneous rocks are classified into four (4) based on their silica content as shown below.

Table 4.4: Rock designation based on silica (SiO_2) content. After Hyndman (1985)

Chemistry	Designation	Rock type
32%-45% SiO_2	Ultra-basic	Komatite
45%-52% SiO_2	Basic Basalt	Gabbro
52%-66% SiO_2	Intermediate	Andesite, Dacite, Phonolite, Tonolite
66%-75% SiO_2	Acidic	Rhyolite, Granite

Based on the SiO_2 content of the table above, the rocks of the Bongwo hills (Three sisters' rock) are acidic granite. Most of the rocks are Calc-alkaline in nature like most other Pan-African granite in Nigeria.

4.4.2.2 Alumina (Al_2O_3) content

Table 4.5: Rock designation based on Alumina (Al_2O_3) content. After Hyndman (1985)

Chemistry	Designation	Distinctive minerals
$\text{Al}_2\text{O}_3 > \text{Na}_2\text{O} + \text{K}_2\text{O} + \text{CaO}$	Peraluminous	Muscovite, Biotite, Corundum, Topaz, Garnet, Tourmaline.
$\text{Al}_2\text{O}_3 < \text{Na}_2\text{O} + \text{K}_2\text{O} + \text{CaO}$ but $\text{Al}_2\text{O}_3 > \text{Na}_2\text{O} + \text{K}_2\text{O}$	Metaluminous	Hornblende, Epidote, Melilite, Biotite Pyroxene.
$\text{Al}_2\text{O}_3 = \text{Na}_2\text{O} + \text{K}_2\text{O}$	Subaluminous	Olivine, Orthopyroxene, Clinopyroxene.
$\text{Al}_2\text{O}_3 < \text{Na}_2\text{O} + \text{K}_2\text{O}$	Peralkaline	Sodic pyroxene, Sodic amphiboles, Astrophyllite, Columbite, Phyllochlorite.

The alkalis Na_2O and K_2O have an average of 33.04% and 75.5% respectively, in total Na_2O and K_2O is 108.54, which is greater than the average of alumina content of the rock (39.8%). These provide for designation of Bongwo Hills (Three Sister Rock) granites as Peralkaline. (Table 4.5)

The Magnesium Oxide (MgO) is very low in all the samples analyzed. The average value of MgO is 16.04%, which is clear indication that the rock samples are acidic.

The percentage of SiO_2 and Al_2O_3 indicate source of magma. However, Bongwo Hills (Three Sister Rock) seem to originate from lithosphere because they compose of Silica (SiO_2) and Alumina (Al_2O_3) or because of fractional crystallization of early formed minerals, which tend to separate from the silicate melt and settle at the bottom by gravity settling thereby enhancing the Silica and Alumina content of the melt. Most of the granite of Bongwo Hills (Three Sister Rock) is of igneous origin. This reflects very well with those obtained from other parts of the Nigerian basement complex.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY

This study examined the geology and petrographic characteristics of the Granite–Gneiss complex in Bongwo Hills (Three Sisters' Rock), Song, Northeastern Nigeria. Detailed geological mapping, petrographic analysis, and geochemical studies were carried out to understand the mineral composition, texture, and origin of the rocks in the area.

Field observations revealed that the major rock types include phaneritic granite, coarse to medium-grained biotite granite, gneiss, migmatite, basalt, and minor pumice. These rocks occur as massive outcrops, boulder ridges, and domes with visible jointing and foliation patterns trending mainly NE–SW, which correspond to the regional Pan-African structural trend.

Petrographic analysis showed that the granitic rocks are mainly composed of quartz, plagioclase, and K-feldspar, with minor biotite, muscovite, and opaque minerals. The texture is holocrystalline and equigranular, confirming that the rocks crystallized slowly from magma at deep crustal levels. The presence of gneiss and migmatite indicates high-grade metamorphism and partial melting, which later modified the original igneous rocks.

Geochemical results from Atomic Absorption Spectrophotometry (AAS) show that SiO_2 and Al_2O_3 are the dominant oxides, with relatively low MgO and moderate alkali contents ($\text{Na}_2\text{O} + \text{K}_2\text{O}$). The Harker variation diagrams reveal systematic trends consistent with fractional crystallization and magmatic differentiation. Based on alumina saturation, the rocks are classified as peralkaline granites.

1.2 CONCLUSION

The integration of field, petrographic, and geochemical data indicates that the rocks of Bongwo Hills belong to the Precambrian Basement Complex of Northeastern Nigeria. The granitic rocks are acidic in composition, and their mineral assemblages suggest crystallization from a silica-rich magma that underwent fractional crystallization. The occurrence of gneiss and migmatite, together with the structural features observed in the field, suggests that these rocks were later subjected to metamorphism and deformation during the Pan-African orogenic event.

In conclusion, the Bongwo Hill (Three Sisters' Rocks) Granitoids are igneous in origin, intruded, and emplaced within the Nigerian Basement Complex.

1.3 RECOMMENDATIONS

1. Further Geochemical Analysis: Additional major-, and trace-element analyses (e.g., XRF, ICP-MS) should be conducted to complement AAS results and provide a more detailed geochemical understanding of the rocks.
2. Geochronological Studies: Radiometric dating (e.g., U–Pb Zircon) is recommended to constrain the emplacement ages of the granites and correlate them with Pan–African events in other parts of the Nigerian Basement Complex.
3. Economic Potential: Detailed mineralogical and geochemical studies of pegmatite, and accessory minerals should be carried out to evaluate the potential for rare earth and industrial mineral exploitation.
4. Hydro Geological Assessment: Given the significance of groundwater in the area, a detailed hydro geological study is advised to quantify aquifer characteristics and sustainable water supply for the community.

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